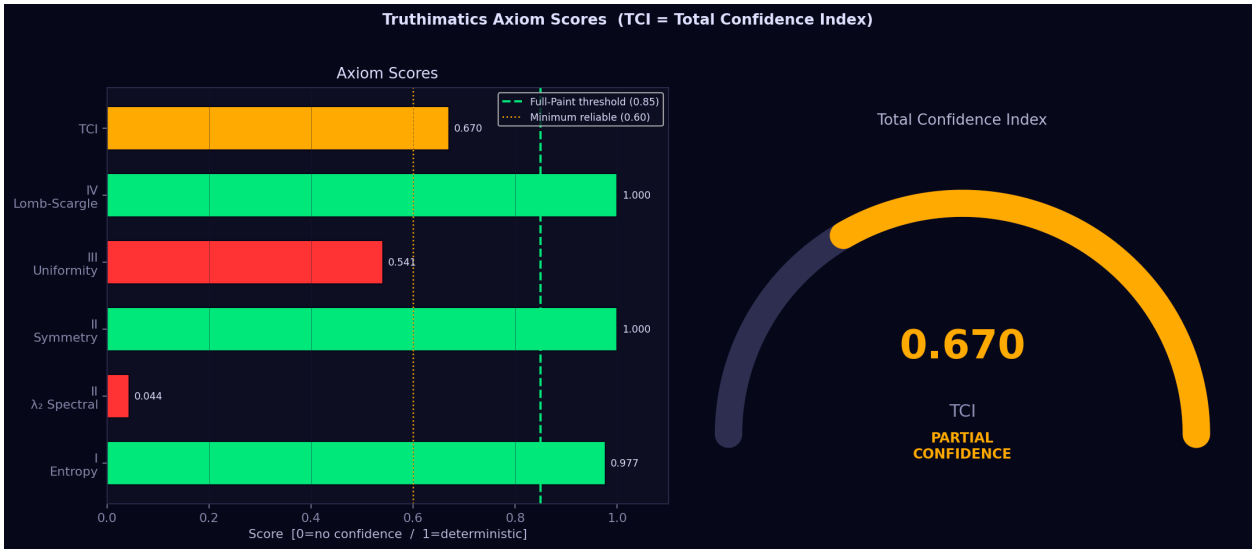
AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_BC5CBCA03A
Taxa	Parasaurolophus
Clade	Ornithischian
PBDB Records	20
Mass (regression)	3791.2 kg (Campione & Evans 2012)
Mass (volumetric)	3963.6 kg (Silhouette integral, ρ=1060.0 kg/m^3)
Body Volume	4.172 m^3
Validity Score	1.0000 / 1.0000
TCI	0.6699 (partial)
Axiom I Entropy	0.9771
Axiom II λ■	0.1763 (symmetry 100%)
Axiom III Uniformity	0.5410
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	56.0% Measured / 44.0% Inferred
Elements	25
CoG (m)	[0.371, 1.055, 0.0]
Generated	2026-05-01 00:01 UTC

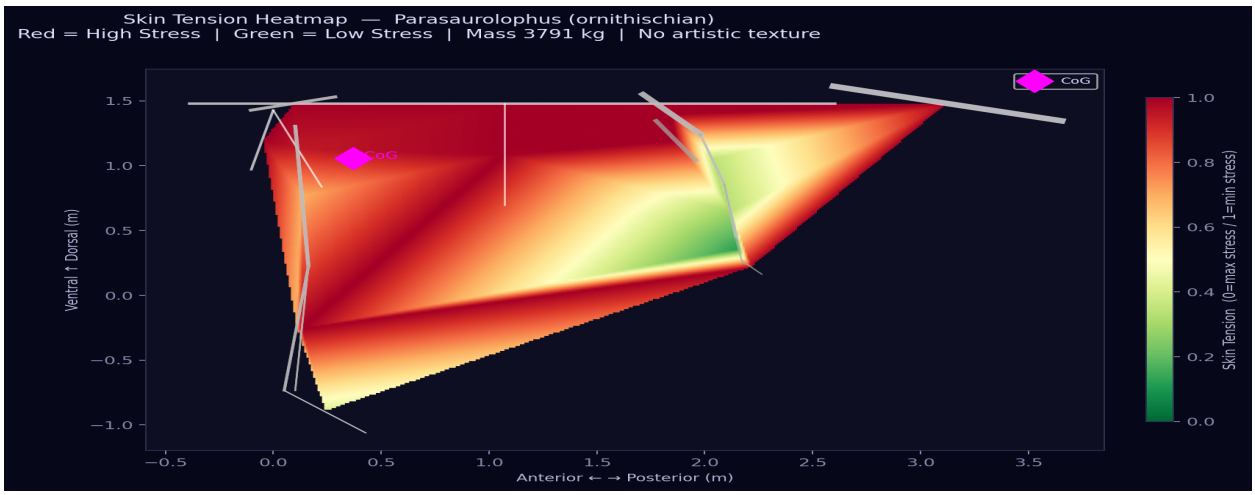
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



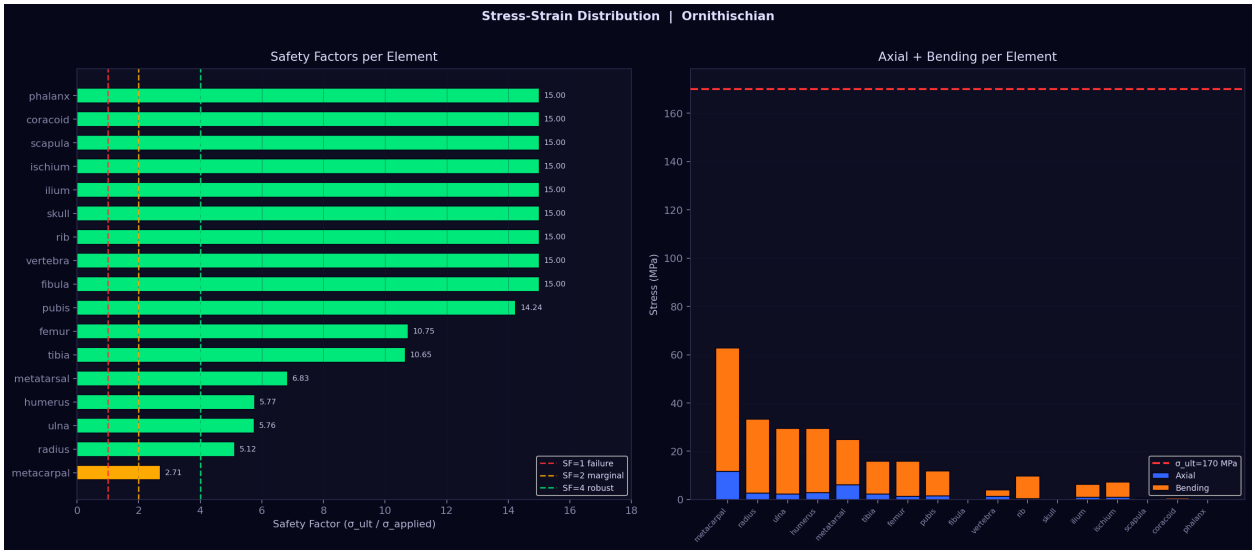
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: quadruped (F=Mg/4). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

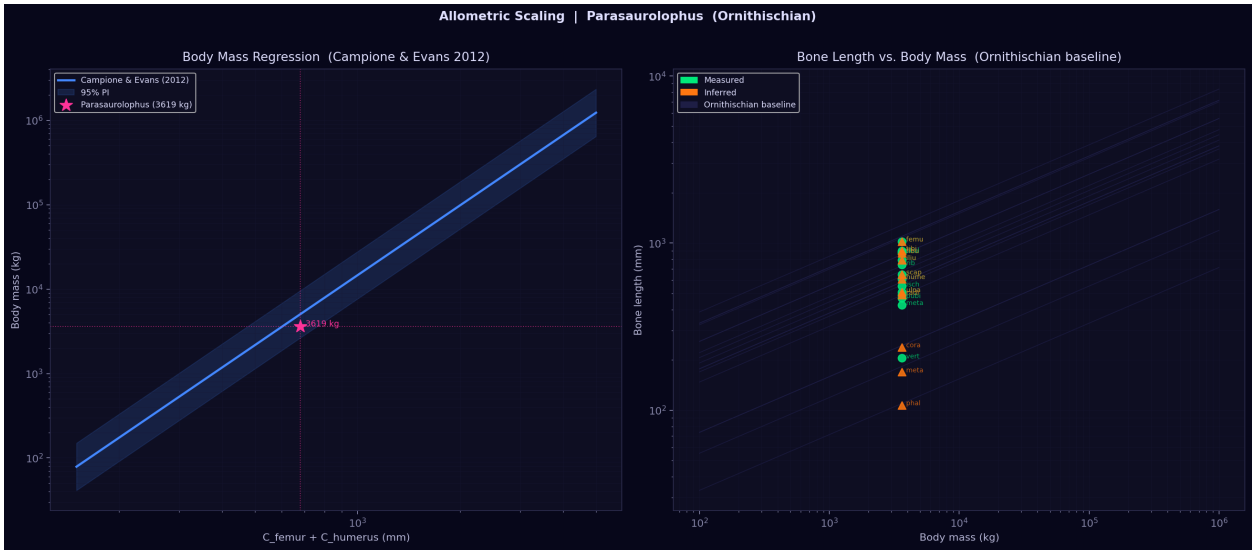


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.253	0.463	0.716	15.000	OK
femur	1.352	14.465	15.818	10.747	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	2.861	26.622	29.483	5.766	OK
ilium	0.959	5.366	6.325	15.000	OK
ischium	0.957	6.306	7.263	15.000	OK
metacarpal	11.605	51.197	62.802	2.707	WARNING
metatarsal	6.155	18.726	24.881	6.833	OK
phalanx	0.000	0.000	0.000	15.000	OK
pubis	1.496	10.442	11.937	14.241	OK
radius	2.564	30.657	33.221	5.117	OK
rib	0.285	9.434	9.719	15.000	OK
scapula	0.137	0.500	0.637	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	2.307	13.651	15.958	10.653	OK
ulna	2.305	27.209	29.514	5.760	OK
vertebra	1.272	2.651	3.923	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	2.345 m	Femur + tibia + metatarsal
Estimated Speed	14.3 km/h (3.97 m/s)	Gait: Walk | Fr=0.684
Stride Length	4.8 m	Alexander (1976) / Thulborn (1990)
Bite Force	16466.6 N (1678.6 kgf)	Ornithischian regression (Norman & Weishampel 1991)
Skull Length	880.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.3942	Circumference / length (dimensionless)
Limb Strength Index	4.2e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	9.17 m	Allometric from femur length
Est. Neck Length	1.01 m	Allometric from femur length
Est. Tail Length	2.57 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	238.5	175.1	2927.15	3654.0	Constraint	0.40	15.00
femur	1020.0	128.0	6693.91	8356.0	Measured	0.88	10.75
femur_right	1020.0	128.0	6693.91	8356.0	Symmetry	0.84	10.75
fibula	872.0	40.0	394.48	413.7	Measured	0.88	15.00
fibula_right	872.0	40.0	394.48	413.7	Symmetry	0.84	15.00
humerus	610.0	88.0	1892.15	2362.0	Measured	0.88	5.77
humerus_right	610.0	88.0	1892.15	2362.0	Symmetry	0.84	5.77
ilium	792.0	152.0	7329.47	9149.4	Measured	0.88	15.00
ilium_right	792.0	152.0	7329.47	9149.4	Symmetry	0.84	15.00
ischium	552.0	90.0	1790.95	2235.6	Measured	0.88	15.00
metacarpal	169.1	42.4	45.26	37.2	Constraint	0.65	2.71
metatarsal	425.0	60.0	612.85	765.0	Measured	0.88	6.83
phalanx	107.3	35.0	15.87	12.3	Constraint	0.40	15.00
pubis	468.0	72.0	971.79	1213.1	Measured	0.88	14.24
radius	490.0	55.0	593.72	741.1	Measured	0.88	5.12
radius_right	490.0	55.0	593.72	741.1	Symmetry	0.84	5.12
rib	740.0	36.0	384.15	479.5	Measured	0.88	15.00
scapula	648.0	238.0	14702.43	18353.0	Measured	0.88	15.00
scapula_right	648.0	238.0	14702.43	18353.0	Symmetry	0.84	15.00
skull	880.0	395.0	54996.74	68652.4	Measured	0.88	15.00
tibia	900.0	98.0	3462.22	4321.9	Measured	0.88	10.65
tibia_right	900.0	98.0	3462.22	4321.9	Symmetry	0.84	10.65
ulna	510.0	58.0	687.20	857.8	Measured	0.88	5.76
ulna_right	510.0	58.0	687.20	857.8	Symmetry	0.84	5.76
vertebra	205.0	132.0	1430.74	1786.0	Measured	0.88	15.00